

1. Consider the work done by the vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ on a particle moving counterclockwise around the unit circle.

- a) Use Green's Theorem to evaluate the resulting line integral.

- b) How would your answer change if the particle was moving clockwise around the unit circle?

- $(2, 0)$, $(0, 2)$, and $(-2, 0)$ in the counterclockwise direction.

3. Evaluate $\oint_C (x^2 + y^2) dx + x dy$, where C is the circle $x^2 + y^2 = 9$.

4. Suppose an object is being spun by high wind currents modelled by the vector field

$$\mathbf{F}(x, y) = \left\langle -y^3 + x \tan^{-1}(y), x^3 + 2 + \frac{x^2}{2 + 2y^2} \right\rangle \text{ in a counterclockwise fashion around a circle of radius 2.}$$

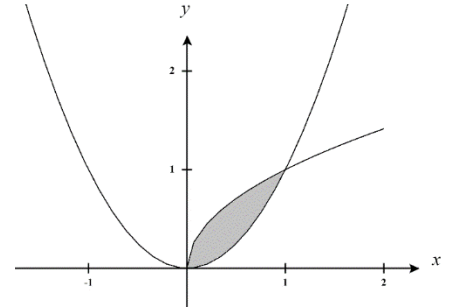
Determine the total work done by the wind currents on the object as the object moves around the circle once.

5. Determine the work done by the vector field $\mathbf{F}(x, y) = \langle e^{\cosh(x)} + 3y, e^{\sinh(y)} - 3x \rangle$ on a particle moving along C , where C is the boundary of the region outside the unit square but inside the circle $4x^2 + 4y^2 = 16$ with positive orientation.

6. Use Green's Theorem to evaluate the line integral along the given positively oriented curve.

$$\int_C (2x + \cos(y^2)) dy + (y + e^{\sqrt{x}}) dx$$

where C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.



7. Explain why green's theorem cannot be used to evaluate the following line integral:

$$\int_C \ln(1+2x) dx + \frac{3}{2} \left(x - \frac{1}{2} \right)^{2/3} dx \quad \text{where } C \text{ is the boundary of positively oriented unit circle.}$$

8. Compute $\int_C xe^{-2x}dx + (x^4 + 2x^2y^2 + y^4)dy$, where C is the negatively oriented boundary of the region enclosed by $y = x^2$, the x -axis, and $x = 2$.

9. Use Green's Theorem to evaluate the following line integral where C is the path from $(1,0)$ to $(3,0)$ along a line, followed by the path from $(3,0)$ to $(-3,0)$ along the top half of a circle of radius 3 centered at the origin, followed by the path from $(-3,0)$ to $(-1,0)$ along a line, followed by the path from $(-1,0)$ to $(1,0)$ along the top half of a circle of radius 1 centered at the origin.

$$\int_C (\tan^{-1}(x) + y^2)dx + (e^y - x^2)dy$$

10. Evaluate $\oint_C \left(x^{\sin(x)} - x - 2y \right) dx + \left(2x - \tan^{-1}(y) - y^{\cos(y)} \right) dy$, where C is the ellipse $x^2 + 4y^2 = 16$.

